

Exploring a Single Autoregressive LLM for Unified Target Speech Extraction across Synchronous and Asynchronous Cues

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Abstract—Target speech extraction (TSE) typically trains a separate extractor per cue, and visual-cue systems often need corruption-matched training to remain robust under visual frame corruption. We show that one autoregressive LLM backbone, TSE-Omni, can serve both temporally synchronous cues (lip movements, co-speech gestures) and asynchronous cues (enrollment audio, text). TSE-Omni is driven by next-token prediction: each step predicts target speech semantic tokens from its own past outputs, which we term self-enrollment, forming a continuous target-speech context initialized by the enrollment cue (asynchronous audio or text, or a short visual prefix). This enables audio-visual compensation: the model uses synchronized visuals when intact and its token history when visual frames are missing. Under clean visuals, TSE-Omni matches strong discriminative and generative baselines (SpeechBERTScore 0.81 on VoxCeleb2 and 0.89 on LRS3 zero-shot) with higher DNSMOS. On the same VoxCeleb2 test set, after a 2s clean visual start, removing the remaining visual frames leaves SpeechBERTScore at 0.81. It remains usable under sparse overlap and multi-speaker interference, and supports streaming inference. Project page: <https://alexwxwu.github.io/tseomni-main/>.

Index Terms—Target speech extraction, audio-visual, large language model, next-token prediction, self-enrollment.

I. INTRODUCTION

Target speech extraction (TSE) aims to extract the speech of a target speaker from a mixture of multiple talkers, based on a cue that indicates *who* to extract, as illustrated in Fig. 1. Cues fall into two categories. *Temporally synchronous* cues, such as lip movements and co-speech gestures, are aligned frame-by-frame with the mixture and can track the target speaker over time. *Temporally asynchronous* cues, such as a pre-recorded audio clip or a text description, are not time-aligned; they help identify the target but cannot track it continuously.



Fig. 1: Cue types. Cues are either *temporally synchronous* with the mixture (lip movements, co-speech gestures) or *temporally asynchronous* (pre-recorded audio, text).

Audio- and visual-cue-based methods are the most widely studied paradigms. Audio-based approaches, such as SpEx [1], usually guide separation with a short pre-recorded enrollment audio, whereas visual-based methods, such as TDSE [2], rely on synchronous lip movements. Recent work provides richer

context: USEF [3] and SoloSpeech [4] apply cross-attention between the audio cue and the mixture speech; AVHuMAR [5] and C^2 AVTSE [6] integrate audio-visual context with a mask-and-recovery strategy; and MeMo [7] builds memory banks from historical speech context for online extraction. Despite this progress, most TSE systems are still optimized mainly at the signal level, without leveraging the phonetic and linguistic knowledge that speech carries.

Meanwhile, large language models (LLMs) have been introduced to TSE. ELEGANCE [8], [9] injects textual linguistic guidance into TSE models. This textual alignment improves the semantic consistency between the extracted speech and the ground truth, and also helps compensate for corrupted visual cues at no extra inference cost. However, its extraction is still not purely semantic-driven.

Some studies use LLMs as speech extractors. LauraTSE [10], built on LauraGPT [11], predicts the first two RVQ layers conditioned on the audio cue and the mixture embeddings in an AR stage, then predicts the remaining codebooks in a NAR stage. LLaSE-G1 [12] unifies speech and audio enhancement with a single-layer codec. These models move from signal-level processing toward semantic-driven extraction, but disentangling the target speech from the mixture remains their bottleneck and the main source of content and acoustic hallucination. Extending them to multiple cues under limited parameters and training data is therefore difficult. Their autoregressive formulation, however, offers a natural remedy: during decoding, an AR model accumulates a history of its own predicted target-speech tokens, which can be reused as a continuously refreshed enrollment cue at no extra cost. We term this mechanism *self-enrollment* (Fig. 2); it supplies speaker identity and linguistic context that persist even when an external cue degrades.

However, existing AR-LLM TSE systems remain confined to a single cue: they accept either an audio cue or a visual cue, but not both, and are trained as separate extractors, so the AR token history is never reused to compensate a missing visual stream. Such single-cue systems fail in two common scenarios. When later visual frames are corrupted or missing (e.g., the target moves out of view), a purely visual model has no fallback and often needs corruption-matched training [13] or a dedicated recovery module [14]. When the target switches mid-mixture, an audio-only model must condition on two audio cues at once, which is ambiguous, and errors made at the switch point accumulate over time.

To address both failure modes and unify different cues within a single model, this study proposes TSE-Omni, one AR-LLM backbone that handles both synchronous and asynchronous cues. It predicts target speech tokens from its own

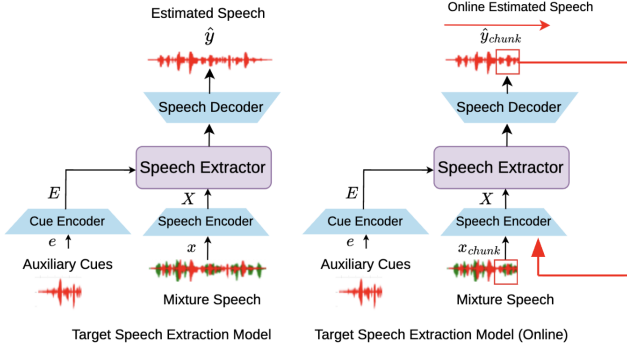


Fig. 2: Self-enrollment compensation: historical predicted target-speech semantic tokens are fed back as a running enrollment cue. This self-loop is inherent to AR decoding. TSE-Omni is the first to exploit it to compensate for corrupted visual cues without a corruption-matched training stage in AR LLM framework.

history (self-enrollment), fusing them with synchronized visual tokens when available, so it can compensate for corrupted or missing visual frames without corruption-matched training, and can follow a target switch mid-mixture. Our main contributions are as follows:

- We propose TSE-Omni, which supports temporally synchronous and asynchronous cues with one AR-LLM extractor backbone, rather than a separately trained extractor per cue.
- We introduce self-enrollment (Fig. 2): the model conditions on its own accumulated speech tokens, so zero-filled corrupt or missing frames (a case seen in omni training) are compensated from the speech-token history, with no corruption-matched training and no recovery module.
- Experiments on VoxCeleb2, plus zero-shot LRS3, visual corruption, visual-guided target switching, sparse overlap, multi-interferer mixtures, and streaming, show that TSE-Omni matches strong baselines on clean-visual semantics while improving perceptual quality, and that SpeechBERTScore is unchanged on the core test set when later visual frames are zero-filled.

II. RELATED WORK

A. Audio-Based and Visual-Based TSE Models

1) *Audio-Based TSE Models*: These models primarily depend on the target speaker’s voice characteristics. Typically, a short pre-recorded speech clip (a few seconds in duration) is used as the enrollment cue. A speaker encoder extracts a speaker embedding from this clip, which subsequently guides extraction of the target speech from the mixture signal, as in SpEx [1]. More recent studies go beyond speaker identity and incorporate broader contextual information from the cue speech. For example, instead of relying solely on speaker encoders, USEF [3] and SoloSpeech [4] employ a cross-attention mechanism between the cue and mixture speech to capture contextual information and reduce latent feature mismatch.

2) *Visual-Based TSE Models*: These models leverage temporally synchronized visual cues, such as lip movements. Typically, a visual speech recognition (VSR) front-end is employed to extract visemes from the target speaker’s lip movements, which are then aligned with the target speaker’s phonemes in the mixture speech, as in TDSE [2]. However, visual corruption frequently occurs in practice and degrades performance. To mitigate this, methods such as ImagineNET [14] adopt an interleaved extractor, where intermediate speech refines corrupted visual cues, and the refined visual cues in turn enhance speech extraction. This iterative design proves effective under severe visual degradation.

Existing audio- and visual-based extractors are developed and deployed independently, so a separate backbone is stored for each cue. This increases deployment cost and makes it harder to reuse complementary cross-modal information at inference. TSE-Omni instead shares one AR-LLM extractor across audio- and visual-cue TSE, with a modality-specific front-end for each cue.

B. Target Speech Extraction with LLMs

Large language models (LLMs) have shown strong capabilities in speech enhancement, separation, and extraction. Current LLM-based TSE methods follow two main paradigms.

The first approach uses LLMs as knowledge bases to provide linguistic cues. For instance, studies such as [8], [9] extract textual linguistic guidance, including linguistic priors, constraints, and predictions, from models like RoBERTa [15] and Qwen3 [16], and inject this information into traditional TSE pipelines. This textual guidance helps compensate for corrupted visual cues, thereby improving semantic coherence and speech quality.

The second approach fine-tunes LLMs directly for speech enhancement or TSE tasks [17], [18]. For example, GenSE [17] employs an AR-LLM to predict clean speech semantic tokens from noisy inputs, and subsequently produces enhanced acoustic tokens conditioned on both original and predicted representations using another AR-LLM. Similarly, LauraTSE [10] adopts a two-stage framework based on LauraGPT [11]. LLaSE-G1 [12] jointly models semantics and acoustics within a single-stage pipeline, utilizing a Llama-style LLM backbone with a single-layer codec (X-Codec2 [19], [20]) under a non-autoregressive strategy. These methods demonstrate the effectiveness of the next-token prediction (NTP) paradigm in speech processing. Despite these advances, existing LLM-based TSE methods remain restricted to audio-only scenarios. To the best of our knowledge, TSE-Omni is the first LLM-based TSE system that accepts visual cues as well as audio cues in one extractor backbone.

C. Omni-LLMs

Recently, multimodal LLMs (MLLMs) have demonstrated strong performance across various speech understanding and generation tasks. Qwen-Omni [21], [22] employs a thinker-talker architecture supporting flexible input modalities, while LongCat-Omni [23] and OmniVinci [24] utilize efficient cross-modal interaction mechanisms for modalities such as speech,

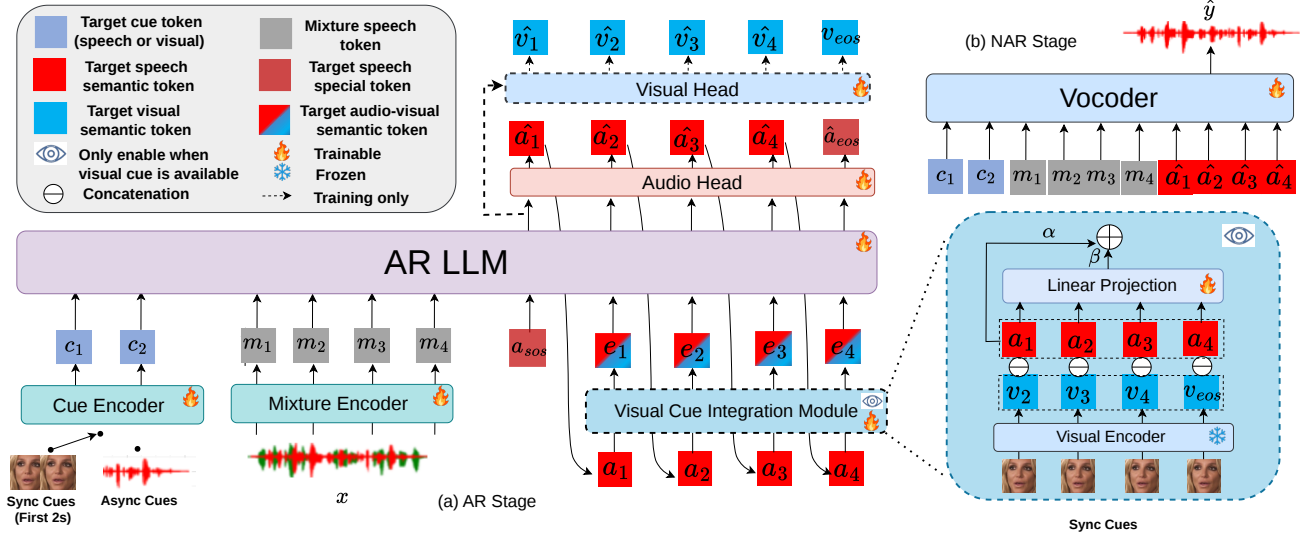


Fig. 3: Overview of the TSE-Omni training framework. (a) **AR Stage**: predicts target speech semantic tokens from the mixture, conditioned on a pre-recorded audio clip or temporally synchronized visual cues. A short initial visual segment (default 2 s) goes to the cue encoder for recognition. The visual integration module is used only for visual-cue TSE. After a_{sos} , visual tokens start at v_2 : each $\hat{a}_t$ ($t \geq 2$) is conditioned on $e_{t-1} = f_{av}(\hat{a}_{t-1}, v_t)$. (b) **NAR Stage**: reconstructs the waveform from the predicted semantic tokens, the mixture, and the cue.

text, and video. These Omni-style MLLMs leverage the foundational understanding and generative capabilities of LLMs to unify diverse single-modality tasks, yielding significant performance gains. Inspired by these advances, we propose TSE-Omni, which extends LLM-based TSE to accept audio and visual enrollment cues, targeting extraction in complex real-world scenarios.

III. METHODS

A. TSE-Omni Overview

Traditional TSE models handle audio and visual cues independently: $\hat{y} = f_a(a, x)$ and $\hat{y} = f_v(v, x)$, where f_a and f_v denote audio- and visual-cue models. This leads to redundant deployment and no parameter sharing. TSE-Omni instead unifies both cues in a single AR-LLM (Fig. 3), consisting of an autoregressive (AR) stage that predicts target speech semantic tokens and a non-autoregressive (NAR) stage that generates the waveform.

AR Stage: The AR stage predicts discrete target speech semantic tokens $\hat{a}_{1:T}$, taking the cue embedding c and the mixture embedding m as context, starting from the special token a_{sos} . The token-wise decoding condition depends on the cue type and is detailed in Sec. III-C, which also introduces our self-enrollment and cross-modality compensation mechanism.

NAR Stage: The NAR stage reconstructs the waveform y from $\hat{a}_{1:T}$, conditioned on m and c . Following [10], [11], a six-layer Conformer predicts the remaining acoustic tokens beyond the first n_q RVQ layers, and a codec decoder produces the speech signal. Both the NAR Conformer and the codec decoder are trained.

B. Mixture and Cue Encoders

As shown in Fig. 3, the LLM’s input at each AR step is a continuous input sequence built from three continuous embeddings: the mixture embedding m , the cue embedding c , and the embeddings of the previously predicted speech tokens $\hat{a}_{1:t-1}$ (looked up from the first n_q RVQ layers).

Mixture Encoder. A six-layer Conformer encoder processes the mel-spectrogram of x (the channel size is first projected to D) and produces the mixture embedding m .

Cue Encoder. The figure shows a single *cue encoder* block, which abstracts a family of modality-specific encoders, one per cue type. Each front-end produces a modality-specific feature, which is then projected to the same D -dimensional space and encoded into a unified continuous cue embedding c :

- **Audio:** a pre-recorded target speech utterance is converted to a mel-spectrogram and encoded (sharing the mixture encoder weights).
- **Visual (Lip movement):** lip embeddings from the default VSR front-end are projected to D (AV-HuBERT is compared in Sec. VI-E3).
- **Visual (Co-speech gesture):** a BLSTM encoder [25] extracts gesture sequence embeddings (15 Hz), which are then upsampled to the speech token frame rate (25 Hz) by interpolation.
- **Text:** RoBERTa [15] extracts utterance-level text embeddings from partial target speech transcripts.

When multiple cues are given, their embeddings are concatenated and projected to the shared space; missing modalities are replaced by zero tensors.

Two Cue Roles. A cue is either *asynchronous* (pre-recorded audio, text) or *synchronous* (lip, gesture, temporally aligned

with the mixture) (Fig. 1). This determines the role a cue plays during extraction:

- **Target-Speaker Recognition.** All cues perform recognition: they identify *who* to extract at the cold start. Asynchronous cues are used directly; for synchronous cues, we use only a short initial segment (e.g., the first 2 seconds), which plays the same recognition role.
- **Target-Speaker Tracking.** Only synchronous cues can track, because they are temporally aligned with the mixture speech and can be fused with each newly predicted speech token $\hat{a}_{t-1}$ (from $t = 2$ onward). For asynchronous cues, tracking is not possible via the cue; instead, it relies solely on the previously predicted speech tokens $\hat{a}_{1:t-1}$ via next-token prediction, as detailed in Sec. III-C.

C. Self-enrollment and Cross-modality Compensation

This is the core of our contribution: we exploit the LLM’s NTP ability for audio-visual cue compensation. The AR stage autoregressively predicts target speech semantic tokens $\hat{a}_{1:T}$, where t indexes the AR step and $\hat{a}_t$ is the stack of the first n_q RVQ code indices predicted at that step (the corresponding look-up embeddings are in $\mathbb{R}^{n_q \times D}$). Following [10], [11], we predict the first $n_q = 2$ of the 32 RVQ layers.

For *audio-cue* TSE, $\hat{a}_t$ is conditioned only on the historical speech tokens $e_{1:t-1} = \hat{a}_{1:t-1}$; the visual module is inactive. For *visual-cue* TSE, the prediction is conditioned on historical audio-visual joint tokens

$$\begin{aligned} e_{1:t-1} &= f_{av}(\hat{a}_{1:t-1}, v_{2:t}) \\ &= \alpha \hat{a}_{1:t-1} + \beta \text{Linear}(\text{concat}(\hat{a}_{1:t-1}, v_{2:t})), \end{aligned} \quad (1)$$

where the residual module f_{av} in Eq. (1) operates independently on each token pair. Concretely, each integrated token e_{t-1} concatenates the look-up embedding of the last predicted token $\hat{a}_{t-1}$ with the synchronized visual token v_t along the channel dimension, and passes the result through f_{av} . Since $\hat{a}_1$ has no preceding token to fuse with, the visual tokens start at v_2 ; each $\hat{a}_t$ is thus conditioned on e_{t-1} , a fusion of the previous speech token $\hat{a}_{t-1}$ and the visual frame v_t .

The residual design of f_{av} keeps the speech token (weight α) while fusing in the projected audio-visual token (weight β). This mitigates modality imbalance, since the LLM backbone is primarily adapted to speech tokens. At inference, later corrupted or missing visual frames are replaced by zeros; the visual module stays on. The AR stage still conditions on $\hat{a}_{1:t-1}$ (self-enrollment), so the β branch sees $\text{concat}(\hat{a}, 0)$ rather than a learned ignore-visual gate. This is the protocol used in the visual-corruption experiments: no corruption-matched training and no extra recovery module. The residual $\alpha \hat{a}$ path is what keeps extraction going after the 2 s clean visual start.

To signal the end of the sequence, we predict the end-of-sequence token a_{eos} using a zero embedding for the corresponding visual token, avoiding an extra special token.

D. Training

During core Omni-training, half of each batch is assigned to audio-cue TSE and half to visual-cue TSE (lip movements

on VoxCeleb2). Text and co-speech gesture cues are trained later on their own mixtures (Sec. V-A). The default loss, used for all main results, is

$$\mathcal{L} = \mathcal{L}_{CE}(\hat{a}, a) + \mathcal{L}_{MSE}(\hat{a}_{emb}, a_{emb}),$$

where $\hat{a}$ and a are predicted and ground-truth discrete speech semantic tokens, and $\hat{a}_{emb}$ and a_{emb} are predicted and ground-truth full target speech embeddings (this MSE trains the NAR Conformer and codec decoder). An optional visual-head loss $\mathcal{L}_{CE}(\hat{v}, v)$ on discrete visual tokens is studied only in the visual-token ablation of Sec. VI-E3; it is not used in the default model.

IV. IMPLEMENTATION

The target speaker is identified by the cue-encoder branch, which recognizes the target identity, while the tracking branch provides frame-level semantic features. Both the speech and visual tokens share a dimension of $D = 128$ and are sampled at 25 Hz.

Training proceeds in two stages. Before the Omni-training stage, the audio cue encoder and the visual cue encoder (excluding its tracking branch) are first aligned with an MSE loss; skipping this alignment leads to clearly degraded performance when the visual cue is used. During the Omni-training stage, the learning rate starts at 10^{-3} and is halved whenever the validation loss does not improve for 3 consecutive epochs, and training stops after 6 consecutive epochs without improvement.

Lip Movement Encoder and Visual Head. We use a VSR (ResNet) [26] front-end to extract visual semantic tokens; it consists of a 3D convolutional layer (Conv3D) followed by a ResNet18 block, pretrained on LRS3 with the lip-reading task. The default model does not attach a visual prediction head. For the optional auxiliary loss in Sec. VI-E3 only, we follow the data-preprocessing protocol of AV-HuBERT Large [27] to obtain discrete visual tokens, feeding only visual data into AV-HuBERT.

Metrics. We use SpeechBERTScore (SBS) [28] to evaluate the semantic similarity between ground-truth and extracted speech. Speaker similarity (SIM) is measured by the WavLM-based similarity metric [29], while speech quality is evaluated with NISQA [30] and DNSMOS [31]. All metrics are widely adopted in generative SE studies [12], [17]; higher is better. Streaming latency is measured by the real-time factor (RTF).

V. EXPERIMENTAL SETUPS

A. Datasets

Core Dataset. We use VoxCeleb2 [32] as our primary training and in-domain evaluation set: its two-speaker mixtures are more challenging, noisier, and much closer to real-world conditions than LRS3, and it provides audio, visual (lip-movement), and transcript cues for the same utterances. We simulate a two-speaker core dataset with 400,000/2,000/2,000 utterances for training/validation/test, respectively (around 400 h of training data). The scale roughly matches the 460 h of LibriSpeech used to pretrain the LauraTSE backbone. That hour count is a data-budget reference only: on Libri2mix we evaluate TSE-Omni zero-shot, whereas LauraTSE is trained

in-domain. Unless a table names another training set, TSE-Omni numbers use this VoxCeleb2 core checkpoint. For audio-cue-based TSE, we randomly select an utterance from the target speaker as a pre-recorded audio cue. All utterances are randomly clipped between 3 and 6 seconds, and the default cue duration is 2 seconds for both audio and visual cues. The interfering speaker is mixed at a signal-to-interference ratio (SIR) randomly sampled between -5 dB and 5 dB.

LRS3 Evaluation. LRS3 mixtures provide both visual and audio cues. Visual-cue baselines on LRS3 (USEV, AV-Sepformer, AV-Mamba, AVDiffuSS, FlowAVSE) are trained in-domain. Table I and the LRS3 ablations test the VoxCeleb2 core checkpoint on two-speaker LRS3 mixtures simulated with the same duration and SIR protocol: TSE-Omni (V) and TSE-Omni (A) are both zero-shot. We do not train TSE-Omni on LRS3, except for the target-switch fine-tuning set below. The VSR front-end remains the LRS3-pretrained lip-reading encoder described in Implementation.

Libri2mix Evaluation. Libri2mix rows in Table I compare in-domain audio baselines with a zero-shot TSE-Omni. NeMo, LLaSE-G1, LauraTSE, and SoloSpeech are trained on Libri2mix (or the LibriSpeech/Libri2mix setup of the original papers). TSE-Omni (A) is the VoxCeleb2 core checkpoint, not trained on Libri2mix. TSE-Omni (A+Pretrain) is initialized from the official LauraTSE checkpoint and then fine-tuned on VoxCeleb2; it is not a Libri2mix-trained TSE-Omni, but it is not zero-shot either, because LauraTSE saw Libri2mix.

Visual-Cue-Impaired Test Set. We apply three corruption types to the VoxCeleb2 core test set (2,000 utterances): full missing, partial occlusion, and low resolution (Fig. 4). To ensure a stable start, the first 2 seconds of each visual cue are kept clean, and corrupted or missing later frames are replaced by zeros; the visual module stays on. The full-missing protocol zeros every visual frame after 2 s (the target leaving the view). Full-missing and the three-type average are reported in Sec. VI-B on this same 2,000-utterance core test set.

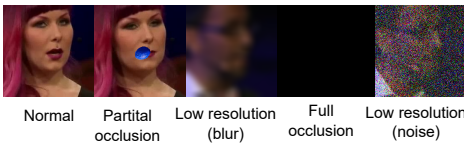


Fig. 4: Visual frame corruption on the VoxCeleb2 evaluation protocol: full missing, partial occlusion, and low resolution. The first 2 s stay clean; later corrupted or missing frames are zeros.

Target-Speaker-Switch Dataset. We randomly select two target utterances of at least 4 seconds (before and after the switch) from LRS3 and simulate 20,000/1,000 mixtures for fine-tuning and test, respectively. In each sample, the target speaker switches mid-utterance at a randomly selected timestamp between 3 and 3.5 seconds. To keep acoustic interference consistent, we randomly select an interfering utterance of at least 6 seconds. This model is fine-tuned from the VoxCeleb2 core checkpoint; it is the only LRS3-trained extractor in

this paper. Switching is evaluated with visual cues (clean or occluded), not with audio-only dual enrollment.

Three-Speaker Test Set. We construct a 3-mix test set from VoxCeleb2 [32] to study the model’s robustness under multi-talker interference. Each mixture contains one target speaker and two interfering speakers, with SIR sampled from $[-10, 10]$ dB. The test set consists of 3,000 utterances. The simulated mixtures have variable durations, randomly between 4–6 seconds. This test set is used only for zero-shot evaluation with the model trained on the core training set.

Sparse-Overlap Dataset. We construct a sparse-overlap dataset from IEMOCAP [33] to study extraction in conversation-style mixture speech: target and interfering utterances are mixed with large silences between the two talkers, while visual inputs are kept clean. The dataset consists of 400,000/9,621/5,528 utterances for training/validation/test. This isolates the model’s ability to predict silent speech tokens from visual cues (e.g., lip non-movement) or pre-enrolled audio cues. The overlap ratio follows USEV [34], with SIR sampled from $[-5, 5]$ dB. The simulated mixtures have variable durations up to 10 seconds, and we use 6-second inputs for inference. This model is fine-tuned from the checkpoint pretrained on the core training set.

Cue-Combination Dataset. To study modality extension and combination, we derive additional cues on VoxCeleb2 [32] and the YGD dataset [25]. VoxCeleb2 provides lip movements, transcripts, and pre-recorded audio; YGD provides co-speech gesture sequences, transcripts, and pre-recorded audio. For both datasets, partial transcriptions are obtained by randomly masking out 20% of the full transcription, extracted with Whisper V2 [35]. For YGD with co-speech gesture cues, we follow the SEG network [25]: we build YGD-2mix from the YouTube gesture dataset [25] of 1,696 TED videos (27,611/3,654/3,475 train/validation/test segments). Following the SEG pipeline, 2D poses are lifted to 3D poses at 15 fps as co-speech gestures; we simulate 200,000/5,000/3,000 mixtures for the training, validation, and test sets at a random interferer SIR in $[-10, 10]$ dB, with 16 kHz audio. VoxCeleb2 cue-combination models are trained from scratch on VoxCeleb2; YGD models are trained from scratch on YGD-2mix.

B. Baselines

We compare TSE-Omni against state-of-the-art baselines, encompassing both generative and discriminative paradigms.

Audio-Cue Baselines: We benchmark against SoloSpeech [4], a DiT-based model with a flow-matching objective; NeMo [36], a generative model utilizing STFT-based spectral processing; LLaSE-G1 [12], a Llama-based TSE model employing discrete tokens and non-autoregressive decoding; and LauraTSE [10], a decoder-only LLM streamlined from LauraGPT [11] for efficient speech extraction.

Visual-Cue Baselines: We include USEV [34] and AV-Sepformer [37], representing RNN-based and Transformer-based architectures respectively, alongside AV-Mamba [9], which leverages state-space models for efficiency. For generative approaches, we compare with AVDiffuSS [38], a

TABLE I: Comprehensive performance benchmarking across LRS3, VoxCeleb2, and Libri2mix. LRS3 supports both visual and audio cues. Visual-cue baselines on LRS3 are trained in-domain; TSE-Omni (V) and TSE-Omni (A) LRS3 rows are zero-shot from the VoxCeleb2 core checkpoint. VoxCeleb2 rows are in-domain. On Libri2mix, audio baselines are trained in-domain; TSE-Omni (A) is zero-shot from the VoxCeleb2 core checkpoint; TSE-Omni (A+Pretrain) is initialized from LauraTSE and fine-tuned on VoxCeleb2. On LRS3 and VoxCeleb2, the best *visual-cue* value per metric is in bold (audio-cue TSE-Omni is listed for reference). On Libri2mix, the best value per metric is in bold. Higher is better. **Param.:** speech extractor backbone parameters; **Arch.:** speech extractor backbone architecture; **Obj.:** prediction objective; **Inp. rep.:** intermediate modeling representation; **Rate:** frame rate; **Type:** D = discriminative, G = generative; **Cue:** A = audio, V = visual.

Test set	Model	Param	Arch.	Obj.	Inp. rep.	Rate	Type	Cue	Semantic-Speaker-Quality [†]			DNSMOS [†]		
									SBS	SIM	NISQA	SIG	BAK	OVL
LRS3	USEV [34]	16.4M	RNN	Mask	Emb	800 Hz	D	V	0.83	0.96	2.95	3.24	3.61	2.80
	AV-Sepf [37]	163M	Sepf.	Mask	Emb	2000 Hz	D	V	0.87	0.96	2.46	3.19	2.90	2.43
	AV-Mamba [9]	126M	Mamba	Mask	Emb	800 Hz	D	V	0.89	0.95	2.82	3.23	3.34	2.66
	AVDiffuSS [38]	241M	U-Net	Noise	STFT	125 Hz	D+G	V	0.65	0.87	3.18	3.35	3.45	2.78
	FlowAVSE [39]	241M	U-Net	Vel.	STFT	125 Hz	D+G	V	0.74	0.93	2.66	3.31	2.89	2.51
	TSE-Omni (V)	77M	AR-LLM	RVQ	RVQ	25 Hz	G	V	0.89	0.96	3.84	3.51	3.76	3.08
	TSE-Omni (A)	77M	AR-LLM	RVQ	RVQ	25 Hz	G	A	0.91	0.97	3.89	3.50	3.79	3.08
VoxCeleb2	USEV [34]	16.4M	RNN	Mask	Emb	800 Hz	D	V	0.80	0.94	2.55	3.18	3.29	2.60
	AV-Sepf [37]	163M	Sepf.	Mask	Emb	2000 Hz	D	V	0.81	0.95	2.08	3.15	2.65	2.29
	AV-Mamba [9]	126M	Mamba	Mask	Emb	800 Hz	D	V	0.79	0.90	2.50	3.17	2.91	2.43
	AVDiffuSS [38]	241M	U-Net	Noise	STFT	125 Hz	D+G	V	0.60	0.74	2.89	3.21	2.89	2.46
	FlowAVSE [39]	241M	U-Net	Vel.	STFT	125 Hz	D+G	V	0.62	0.76	2.69	3.11	2.96	2.46
	TSE-Omni (V)	77M	AR-LLM	RVQ	RVQ	25 Hz	G	V	0.81	0.94	3.24	3.48	3.35	2.85
	TSE-Omni (A)	77M	AR-LLM	RVQ	RVQ	25 Hz	G	A	0.79	0.93	3.23	3.47	3.33	2.84
Libri2mix	NeMo [36]	1.7B	U-DIT	Vel.	STFT	125 Hz	G	A	0.53	0.85	2.21	3.29	3.13	2.61
	LLaSE-G1* [12]	3.2B	AR-LLM*	VQ	Emb	50 Hz	G	A	0.73	0.90	3.89	3.30	3.94	2.99
	LauraTSE [10]	77M	AR-LLM	RVQ	RVQ	25 Hz	G	A	0.90	0.97	4.01	3.60	4.06	3.21
	SoloSpeech fast [†]	513M	U-DIT	Vel.	STFT-VAE	50 Hz	G	A	0.88	0.97	3.94	3.55	4.02	3.25
	TSE-Omni (A)	77M	AR-LLM	RVQ	RVQ	25 Hz	G	A	0.83	0.94	3.69	3.57	4.06	3.29
	TSE-Omni (A+Pretrain)[‡]	77M	AR-LLM	RVQ	RVQ	25 Hz	G	A	0.88	0.97	3.98	3.58	4.05	3.28

*LLaSE-G1 uses non-autoregressive decoding. [†]SoloSpeech fast is the version without the refine stage. [‡]TSE-Omni (A+Pretrain) is initialized from LauraTSE (Libri2mix) and fine-tuned on VoxCeleb2, then tested on Libri2mix. TSE-Omni (A) is zero-shot from the VoxCeleb2 core checkpoint. Libri2mix baselines are trained in-domain.

two-stage pipeline combining discriminative extraction with generative refinement, and with its flow-matching variant FlowAVSE [39].

VI. RESULTS AND ANALYSIS

A. Comparison with Other Backbones

In this section, we evaluate TSE-Omni against widely used audio-cue-based and visual-cue-based TSE baselines on the VoxCeleb2 core test set and, zero-shot, on LRS3 (visual and audio cues) and on Libri2mix (audio only). As shown in Table I, on the LRS3 test set (in-domain visual baselines; TSE-Omni zero-shot from the VoxCeleb2 checkpoint), TSE-Omni (V) achieves an SBS of 0.89, tying the strongest visual-cue baseline (AV-Mamba), while delivering the highest speech quality among all compared models (NISQA 3.84). On VoxCeleb2 (Vox2), our primary evaluation set, TSE-Omni (V) matches the best visual baseline in semantic similarity (SBS 0.81, tied with AV-Sepformer) while attaining markedly higher speech quality (NISQA 3.24, vs. 2.55/2.08/2.50 for USEV/AV-Sepformer/AV-Mamba, respectively). Similar trends hold for speaker similarity (SIM) and the DNSMOS metrics (SIG, BAK, OVL), indicating that TSE-Omni successfully incorporates visual cues into an audio-oriented LLM-TSE architecture while largely retaining its audio-cue capability (Vox2 audio-cue SBS 0.79).

For audio cues on Libri2mix, the baselines are trained in-domain, while TSE-Omni (A) is zero-shot from the VoxCeleb2 checkpoint. TSE-Omni (A) reaches an SBS of 0.83, below LauraTSE (0.90), but still records the best overall quality (OVL 3.29, above LauraTSE’s 3.21). Initializing from LauraTSE and fine-tuning on VoxCeleb2 (TSE-Omni (A+Pretrain)) is no longer zero-shot on this set; it raises SBS from 0.83 to 0.88, closing most of the semantic gap to in-domain LauraTSE while keeping OVL 3.28.

A notable distinction lies in the compression rate of the input speech representations. TSE-Omni employs a 25 Hz frame rate for both speech and visual semantic tokens, whereas most baselines utilize a more fine-grained representation with lower compression rates. This suggests that during the extraction stage, TSE-Omni primarily operates at the semantic level, which makes it reasonable to align speech and visual semantic tokens at this stage, while other baselines place greater emphasis on acoustic details.

B. Speech Token Compensation

The self-enrollment design gives TSE-Omni a unique advantage: since every AR prediction is conditioned on the previously predicted target speech semantic tokens, the model can fall back on this accumulated speech-semantic context to keep tracking the target speaker when later visual frames are

TABLE II: VoxCeleb2 core test set (2,000 utterances), full-missing visual cues: first 2s clean, later visual frames zeros. Same mixtures as the clean VoxCeleb2 test in Table I.

Model	Cue	Semantic-Speaker-Quality $\uparrow$			DNMOS $\uparrow$		
		SBS	SIM	NISQA	SIG	BAK	OVL
USEV [34]	V	0.71	0.86	2.56	3.17	2.94	2.43
AV-Sepf [37]	V	0.70	0.84	1.96	3.04	2.54	2.21
AV-Mamba [9]	V	0.79	0.86	2.42	3.14	2.89	2.39
AVDiffuSS [38]	V	0.60	0.74	2.89	3.21	2.89	2.46
FlowAVSE [39]	V	0.62	0.76	2.69	3.11	2.96	2.46
TSE-Omni (V)	V	0.81	0.95	3.06	3.46	3.32	2.84

zero-filled (Fig. 5). This protocol does not use corruption-matched training data or a recovery module; corrupted frames are zeros, and the visual module stays on.

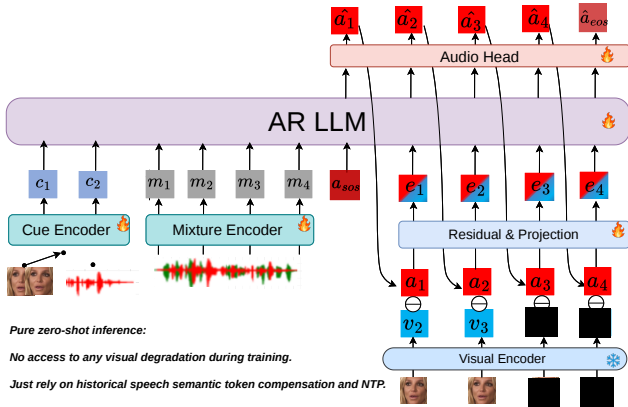


Fig. 5: Speech semantic token compensation. Only the first 2s of visual cues are available; later visual frames are zeros. TSE-Omni continues extraction from the historical predicted speech semantic tokens (self-enrollment), with the visual module still enabled.

We first validate compensation under the most challenging setting: full missing, where only the visual cues of the first 2s are available and later frames are zeros, mimicking a target speaker who moves out of view. As shown in Table II, baselines that rely on continuously available visual cues degrade severely: USEV and AV-Sepformer drop to SBS of 0.71 and 0.70, respectively, and AV-Sepformer’s NISQA collapses to 1.96. In contrast, TSE-Omni (V) maintains an SBS of 0.81, a SIM of 0.95, and a NISQA of 3.06 on this same 2,000-utterance test set as the clean VoxCeleb2 evaluation, confirming that a 2s visual cold start plus speech-token self-enrollment is sufficient to sustain extraction.

We then compare TSE-Omni (V) with USEV under clean and impaired visual conditions on the VoxCeleb2 test set, with results in Fig. 6. Two conclusions emerge. **First**, TSE-Omni (V) consistently outperforms USEV under both conditions and on both metrics: on SBS it reaches 0.810 (clean) and 0.808 (impaired), against USEV’s 0.800 and 0.781; on NISQA the gap is even larger, 3.24 and 3.08 versus 2.55 and 2.18.

Notably, even under impaired conditions, TSE-Omni’s NISQA (3.08) stays well above USEV’s *clean* NISQA (2.55). **Second**, self-enrollment makes TSE-Omni far less sensitive to visual degradation: from clean to impaired, USEV’s NISQA drops by 0.37 and its SBS by 0.019, whereas TSE-Omni degrades by only 0.16 and 0.002, respectively; SBS on the core test set is effectively unchanged when later visual frames are zero-filled, while NISQA still drops (3.24 to 3.08).

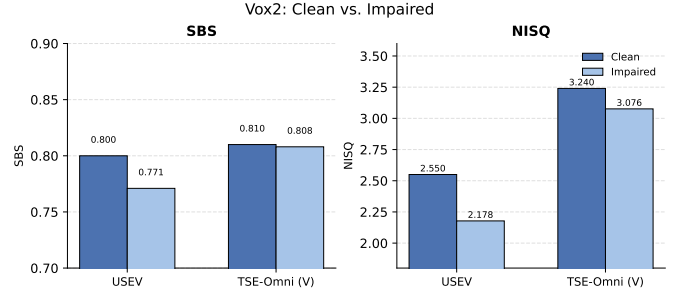


Fig. 6: VoxCeleb2, TSE-Omni (V) vs. USEV: clean visual cues versus the average of full missing, partial occlusion, and low resolution (2s clean start, later frames zero-filled). Metrics are SBS and NISQA.

More importantly, real-world visual degradation is open-ended: occlusion, low resolution, motion blur, poor lighting (Fig. 4), or the speaker leaving the frame cannot be exhaustively anticipated. This work considers three representative types, namely full missing, partial occlusion, and low resolution, and the impaired results in Fig. 6 are their average.

Existing visual-cue TSE models often need corruption-matched training or a recovery module, or they degrade sharply (Table II). TSE-Omni instead zeros later visual frames and relies on self-enrollment; we do not train on simulated corruptions or add a visual-recovery module.

C. Challenging Scenarios

1) *Target Speaker Switch*: We evaluate TSE-Omni on the target speaker switching scenario, where the target speaker changes mid-utterance, under two visual conditions: clean (vclean) and occluded (vocc), as illustrated in Fig. 7. Results are reported in Table III.

TABLE III: Target speaker switch on LRS3 (fine-tuned from the VoxCeleb2 core checkpoint). “SWS” is the switch token. We report DNMOS-OVL only; SIG/BAK follow the same trend. Best values within each visual condition are bold.

Method	Visual	SWS	Sem.-Spk.-Qual $\uparrow$			DNMOS $\uparrow$
			SBS	SIM	NISQA	OVL
USEV	vclean	–	0.808	0.949	2.635	2.744
		w/o	0.774	0.928	3.495	3.082
		w/	0.812	0.936	3.560	3.120
USEV	vocc	–	0.736	0.926	2.477	2.650
		w/o	0.766	0.925	3.474	3.074
		w/	0.809	0.934	3.560	3.140

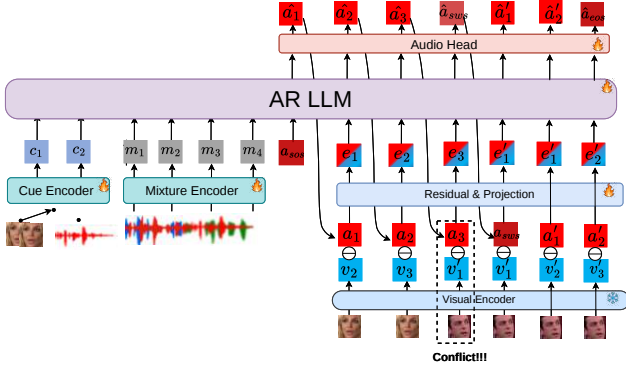


Fig. 7: Target-speaker switch on a visual cue: a switch token (SWS) is added to the vocabulary. The LLM detects a conflict between the new visual cue and the self-enrolled speech history, resets the AR context, and follows the new target.

Two observations emerge from Table III. **First**, TSE-Omni with the switch token (w/ SWS) consistently outperforms USEV on all quality metrics under both conditions, and the advantage is larger under vocc, where visual cues are degraded (SBS 0.809 vs. 0.736). USEV retains the highest SIM under vclean (0.949), since masking-based extraction preserves the timbre of the tracked speaker, but it suffers severe quality degradation (NISQA 2.635). **Second**, incorporating SWS yields consistent gains over its counterpart without the token (w/o SWS): SBS improves from 0.774 to 0.812 under vclean and from 0.766 to 0.809 under vocc (an absolute gain of about 0.04), confirming that the switch token is the key to handling speaker transitions.

These results indicate that the LLM backbone successfully learns to predict the SWS special token at the exact switching point, triggered by a conflict between the new target speaker’s visual cue and the historical speech-token prediction. This mechanism effectively resets the accumulated self-enrollment history and mitigates inter-speaker interference during transitions.

2) *Three-Speaker*: Table IV presents the zero-shot performance comparison on the VoxCeleb2 test set under three-speaker interference (one target, two interferers). TSE-Omni (A) is best on every reported metric. TSE-Omni (V) matches USEV on SBS (0.629 vs. 0.626) and is much higher on quality (NISQA 2.820 vs. 1.774), but its SIM (0.867) is below USEV (0.879). These findings highlight generalization under extra interferers, with audio enrollment remaining the stronger identity cue in this 3-mix setting.

3) *Sparse Overlap*: We evaluate extraction under conversation-style sparse overlap on IEMOCAP [33], where the two talkers alternate with long silences and visual cues are kept clean. As shown in Table V, TSE-Omni (V) and TSE-Omni (A) perform comparably: they are slightly lower than USEV on SBS and SIM, but clearly better on all quality metrics (NISQA 1.52 vs. 1.30; OVL 2.26 vs. 1.98).

This trade-off reflects the different behaviors of generative models on sparse-overlap mixtures: the AR-LLM backbone

TABLE IV: Three-speaker mixtures on the VoxCeleb2 test set (zero-shot from the core checkpoint). Best values within each metric are bold.

Model	Semantic-Speaker-Quality $\uparrow$			DNSMOS $\uparrow$		
	SBS	SIM	NISQA	SIG	BAK	OVL
USEV [34]	0.626	0.879	1.774	3.037	2.327	2.087
TSE-Omni (V)	0.629	0.867	2.820	3.393	2.819	2.513
TSE-Omni (A)	0.658	0.898	2.827	3.395	2.843	2.535

TABLE V: Target speaker extraction under sparse overlap on IEMOCAP. TSE-Omni (V) uses clean visual cues; TSE-Omni (A) uses audio enrollment. Fine-tuned from the VoxCeleb2 core checkpoint. Best values are bold.

Model	Semantic-Speaker-Quality $\uparrow$			DNSMOS $\uparrow$		
	SBS	SIM	NISQA	SIG	BAK	OVL
USEV [34]	0.774	0.872	1.300	2.620	2.600	1.980
TSE-Omni (V)	0.750	0.821	1.520	2.850	2.910	2.260
TSE-Omni (A)	0.743	0.817	1.510	2.840	2.870	2.240

can leverage visual cues (e.g., lip non-movement) or pre-enrolled audio cues to predict silent speech tokens, which occasionally introduces minor false extractions (lower SBS/SIM) but produces cleaner output on NISQA and DNSMOS. We do not claim a semantic win on this set.

4) *Streaming Inference*: MeMo [7] is an online AV-TSE framework tailored for streaming inference. We compare it with TSE-Omni using a chunk size of 1 second, with results in Table VI.

MeMo achieves slightly better SBS and SIM, while TSE-Omni delivers superior speech quality across all metrics (e.g., NISQA 2.73 vs. 2.02 with the LauraGPT backbone). Regarding efficiency, MeMo attains the lowest RTF of 0.03 thanks to its RNN-based computation, but it requires a 2-second warm-up via a pseudo-AR sliding window before extraction can begin. TSE-Omni has no such pseudo-AR warm-up; it still uses the first 2 s of the visual cue for recognition, as in offline visual TSE. Its RTF is 0.64 with the LauraGPT backbone (limited by autoregressive sampling and multi-head attention) and reduces to 0.23 with the 77M Qwen2.5 backbone thanks to grouped-query attention. This latency reduction, however, comes at the cost of a visible semantic-quality drop (SBS 0.75 $\rightarrow$ 0.64 for the audio cue), a trade-off that should be weighed for each deployment scenario.

The performance drop relative to offline inference is likely due to the NAR stage, where the vocoder benefits from a broader context than a single 1-second chunk provides.

D. Potential of Modality Extension and Combination

Fig. 8 shows how cue embeddings are concatenated. Table VII reports Audio, Text, and Visual combinations on VoxCeleb2 and YGD.

Main findings:

- **Combining Audio and Lip Cues Outperforms Either Alone.** Combining audio and lip cues (A + V) gives the

TABLE VI: Streaming on the VoxCeleb2 test set, 1 s chunks. Qwen2.5 is the 77M from-scratch backbone (same size as LauraGPT), not Qwen2.5-0.5B. Best values are bold.

Model	Backbone	Cue	SBS	SIM	NISQA	SIG	BAK	OVL	RTF↓
MeMo [7]	–	V	0.76	0.95	2.02	3.15	2.71	2.35	0.03
TSE-Omni	LauraGPT	A	0.75	0.90	2.73	3.36	3.21	2.73	0.64
		V	0.72	0.89	2.64	3.35	3.19	2.71	0.65
	Qwen2.5	A	0.64	0.73	2.53	3.40	3.09	2.68	0.23
		V	0.65	0.72	2.56	3.38	3.14	2.69	0.24

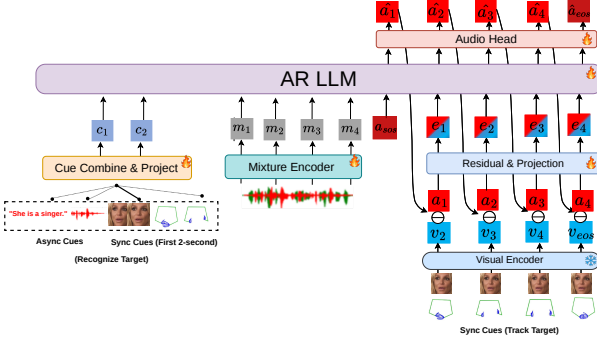


Fig. 8: Multi-cue fusion: each encoder is mapped by a linear layer into the shared D -dimensional space; a missing cue is a zero tensor. Used for Table VII.

TABLE VII: Ablation of cue combinations. VoxCeleb2 columns: trained from scratch on VoxCeleb2 (V = lip movements). YGD columns: trained from scratch on YGD-2mix (V = co-speech gestures). Not the same checkpoint. Bold = best among TSE-Omni rows.

Model	Cue	Vox2			YGD		
		SBS	SIM	OVL	SBS	SIM	OVL
USEV / SEG†	V	0.800	0.940	2.600	0.650	0.840	2.580
TSE-Omni	A	0.790	0.921	2.851	0.713	0.835	2.893
	T	0.602	0.746	2.874	0.442	0.519	2.643
	V	0.803	0.937	2.831	0.602	0.732	2.776
	A+T	0.771	0.925	2.848	0.704	0.844	2.964
	A+V	0.810	0.953	2.852	0.701	0.838	2.772
	T+V	0.772	0.920	2.851	0.700	0.836	2.893
	A+T+V	0.771	0.920	2.851	0.709	0.833	2.837

†Baseline: the conventional visual method per dataset (Vox2: USEV; YGD: SEG), using V; V is lip movements on Vox2 and co-speech gestures on YGD.

highest speaker similarity on Vox2 (**0.953**), consistent with video helping to track the target speaker.

- **Text Cues Perform Poorly.** Text-only (T) results are weak on SBS/SIM (Vox2: 0.602/0.746; YGD: 0.442/0.519), largely because there is not enough training data to properly match textual embeddings, mixture embeddings, and speech semantic tokens. (Its relatively high Vox2 OVL of 2.874 notwithstanding, text-only extraction fails to track the target identity.)
- **Mixing Strong and Weak Modalities Can Hurt.**

Adding a weak modality to a strong setup often lowers overall performance, indicating that multi-modal fusion remains non-trivial.

- **Co-speech gesture cues alone are not sufficient.** On YGD, gesture cues (V) perform worse than lip movements (Vox2’s V) because body language mainly provides prosody and lacks the detailed visemes found in lip movements (SBS 0.602 vs. 0.803). In addition, directly upsampling gestures to match the frame rate of speech semantic tokens is ineffective.

E. Ablation Studies

1) *LLM Backbones and Initialization:* We examine three questions: **Q1**, whether a more advanced LLM architecture helps (LauraGPT vs. Qwen2.5, both trained from scratch); **Q2**, whether pretrained textual weights are beneficial (Qwen2.5-0.5B with vs. without initialization); and **Q3**, whether fine-tuning from the official LauraTSE checkpoint outperforms training from scratch. Results are reported in Table VIII.

TABLE VIII: LLM architecture, textual initialization, and LauraTSE initialization. All rows are trained on VoxCeleb2 (in-domain).

Backbone	Param	Mode	Cue	SBS	SIM	NISQA	OVL
Q1: LLM architecture (from scratch)							
LauraGPT	77M	Scr	A	0.783	0.931	3.230	2.840
			V	0.806	0.941	3.240	2.850
Qwen2.5	77M	Scr	A	0.706	0.900	3.180	2.740
			V	0.710	0.893	3.220	2.770
Q2: Pretrained textual weights (Qwen2.5-0.5B)							
Qwen2.5	0.5B	Scr	A	0.489	0.550	1.873	2.510
			V	0.491	0.550	1.834	2.456
		Init	A	0.667	0.728	2.786	2.857
			V	0.692	0.747	2.796	2.859
Q3: Initialization from LauraTSE							
LauraGPT	77M	Scr	A	0.783	0.931	3.230	2.840
			V	0.806	0.941	3.240	2.850
LauraGPT [†]	77M	Init	A	0.807	0.900	3.090	2.849
			V	0.789	0.902	3.109	2.868

Scr: from scratch; Init: from a pretrained checkpoint (Qwen2.5-0.5B in Q2, LauraTSE in Q3). A/V: audio/visual cue. **Bold:** best per block.

† LauraTSE checkpoint (LibriSpeech/Libri2Mix), fine-tuned on VoxCeleb2.

Three observations can be drawn from Table VIII. (1) With a comparable parameter budget (77M) and training recipe, the LauraGPT-style backbone consistently outperforms the

downscaled Qwen2.5 (e.g., visual-cue SBS 0.806 vs. 0.710). Despite similar parameters, Qwen2.5 uses a more sophisticated architecture than the compact LauraGPT decoder; its higher capacity makes it prone to overfitting on the limited training data, so the simpler, speech-adapted LauraGPT backbone generalizes better. **(2)** Trained from scratch, Qwen2.5-0.5B fails to learn the task ($\text{SBS} < 0.50$), whereas initializing with pretrained textual weights lifts SBS to 0.67–0.69, confirming the value of pretrained linguistic knowledge; nevertheless, it still lags behind the 77M LauraGPT backbone under our data budget. **(3)** Initializing from the LauraTSE checkpoint improves audio-cue SBS (0.807 vs. 0.783) but slightly lowers visual-cue SBS and NISQA, which we attribute to the domain gap between LibriSpeech/Libri2mix pretraining and VoxCeleb2 fine-tuning. On Libri2mix, where baselines are in-domain, TSE-Omni (A) is zero-shot ($\text{SBS} 0.83$). TSE-Omni (A+Pretrain) is LauraTSE initialization plus VoxCeleb2 fine-tuning and reaches SBS 0.88 (Table I); that row should not be read as a Libri2mix-trained TSE-Omni.

2) *Audio-Visual Token Integration Strategy: Dual-Path vs. Uni-Path.* We compare two token integration strategies, illustrated in Fig. 9; results are reported in Table IX.

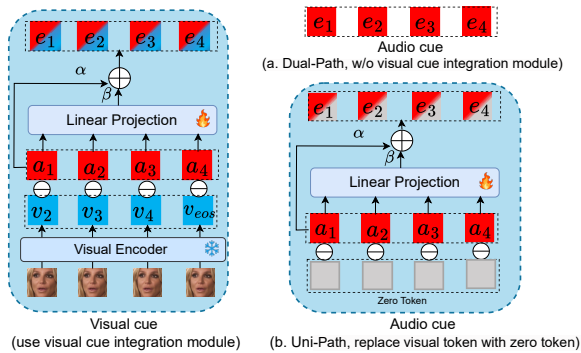


Fig. 9: Token integration. Dual-Path: the visual module is used only with a visual cue. Uni-Path: one module for both cues, with zeros in place of visual tokens for audio-cue TSE.

TABLE IX: Dual-Path vs. Uni-Path token integration on the LRS3 two-speaker test set (zero-shot from the VoxCeleb2 core checkpoint). Best values within each cue are bold.

Cue	Strategy	Semantic-Speaker-Quality $\uparrow$			DNSMOS $\uparrow$		
		SBS	SIM	NISQA	SIG	BAK	OVL
Audio	Dual-Path	0.91	0.97	3.89	3.50	3.79	3.08
	Uni-Path	0.89	0.97	3.72	3.46	3.73	3.05
Visual	Dual-Path	0.89	0.96	3.84	3.51	3.76	3.08
	Uni-Path	0.88	0.95	3.79	3.48	3.75	3.05

Dual-Path consistently outperforms Uni-Path for both cues, with the largest gap in speech quality (NISQA 3.89 vs. 3.72 for audio cues; 3.84 vs. 3.79 for visual cues). Uni-Path forces audio-cue extraction to pass through the visual integration module with zero-padded visual tokens, introducing an unnecessary distribution shift. Dual-Path also avoids feeding zeros during *clean* audio-cue inference. Visual-corruption inference

is different: later visual frames are zeros, but the visual module stays on, which is the protocol of Sec. III-C. We adopt Dual-Path and enable the visual integration module only when a visual cue is used.

Residual Scales. We further investigate the gating scales in the visual integration module, where α weights the historical speech semantic tokens and β weights the fused audio-visual tokens; training and inference use consistent settings. Results are reported in Table X.

TABLE X: Effect of the residual scales on the LRS3 two-speaker test set (zero-shot from the VoxCeleb2 core checkpoint). α : scale of the speech semantic tokens; β : scale of the fused audio-visual tokens. Best values are bold.

α	β	Semantic-Speaker-Quality $\uparrow$			DNSMOS $\uparrow$		
		SBS	SIM	NISQA	SIG	BAK	OVL
0	1	0.79	0.92	3.74	3.33	3.74	2.94
1	1	0.89	0.96	3.84	3.51	3.76	3.08
1	10	0.86	0.96	3.81	3.46	3.72	3.01
10	1	0.88	0.96	3.64	3.43	3.61	2.95

Equal scales ($\alpha = \beta = 1$) give the best performance across all metrics. Setting $\alpha = 0$, i.e., relying solely on fused audio-visual tokens without the speech residual, drops SBS to 0.79, confirming that the speech-semantic stream remains the primary information source for extraction. Conversely, overweighing the self-enrollment branch ($\alpha = 10$) causes the largest quality drop (NISQA 3.64), suggesting that exclusively trusting historical predictions amplifies error accumulation. Overall, the residual connection from historical speech tokens is essential: it is this connection that keeps self-enrollment available when later visual frames are zeros. It still has to be balanced with the visual branch.

3) *Stronger Visual Semantic Tokens:* We explore two ways to strengthen the visual representation: replacing the lightweight VSR (ResNet) encoder with the heavier AV-HuBERT (Large) as the visual encoder, and adding an auxiliary visual head that predicts discrete visual semantic tokens. As shown in Table XI, neither change improves extraction. The stronger encoder yields an SBS virtually identical to ResNet’s, which is consistent with the limited capacity of the 77M backbone to absorb deeper visual representations; adding the visual head slightly lowers SBS. We therefore keep the lightweight ResNet and *omit* the visual head in the default model (Table I). The auxiliary visual loss is reported only here; we do not use it to claim extra viseme–phoneme regularization in the main system.

4) *Cue Robustness:* We further study the cue injection mechanism and answer one question: is it beneficial to enroll the cue at both the AR and NAR stages, or is the AR stage sufficient? Results are summarized in Table XII.

Enrollment stage. Enrolling the cue at both stages consistently outperforms AR-only enrollment across all metrics for both cues. The NAR Conformer and codec decoder are trained with this extra cue input; the gain is therefore not from a frozen vocoder that has never seen the cue.

TABLE XI: Visual modeling strategies on the LRS3 two-speaker test set (zero-shot from the VoxCeleb2 core checkpoint). **In-V Rep.**: input visual representation. **V-Loss**: whether the visual CE loss (and hence the visual head) is enabled. **Δ SBS** is the change in SBS relative to the best value in the table (0.895). All configurations are within ± 0.011 of this best, i.e., differences are statistically negligible. The default model uses VSR (ResNet) without V-Loss.

In-V Rep.	V-Loss	SBS	Δ SBS
VSR (ResNet)	×	0.894	-0.001
VSR (ResNet)	✓	0.887	-0.008
AV-HuBERT (L)	×	0.895	0.000
AV-HuBERT (L)	✓	0.884	-0.011

TABLE XII: Effect of the cue enrollment stage on the LRS3 two-speaker test set (zero-shot from the VoxCeleb2 core checkpoint). Best values within each cue are bold.

Cue	Enr. Stage	Semantic-Speaker-Quality $\uparrow$			DNSMOS $\uparrow$		
		SBS	SIM	NISQA	SIG	BAK	OVL
Audio	AR	0.894	0.967	3.680	3.460	3.740	3.020
	AR+NAR	0.914	0.973	3.890	3.500	3.790	3.080
Visual	AR	0.886	0.960	3.820	3.490	3.740	3.050
	AR+NAR	0.894	0.960	3.840	3.510	3.760	3.080

VII. CONCLUSION

We have presented TSE-Omni, a unified framework that performs both audio- and visual-cue target speech extraction with a single autoregressive LLM, removing per-modality models and reducing deployment cost. TSE-Omni predicts target speech semantic tokens from its own prediction history (self-enrollment), fused with synchronized visual cues via a residual module. This yields cross-modality compensation without corruption-specific training: when visual cues are degraded or missing, extraction continues from the accumulated speech-semantic context, with no visual-recovery module. With few parameters, TSE-Omni matches or surpasses state-of-the-art baselines across most scenarios and clearly leads under visual corruption, speaker switching, and sparse overlap, while supporting streaming inference and extending to further cues such as text and co-speech gestures. Future work will explore stronger LLM backbones, more cue modalities, and lower streaming latency while improving robustness.

REFERENCES

- [1] C. Xu, W. Rao, E. S. Chng, and H. Li, “Spex: Multi-scale time domain speaker extraction network,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 1370–1384, 2020.
- [2] J. Wu, Y. Xu, S.-X. Zhang, L.-W. Chen, M. Yu, L. Xie, and D. Yu, “Time domain audio visual speech separation,” in *2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2019, pp. 667–673.
- [3] B. Zeng and M. Li, “Usef-tse: Universal speaker embedding free target speaker extraction,” *IEEE Transactions on Audio, Speech and Language Processing*, 2025.
- [4] H. Wang, J. Hai, D. Yang, C. Chen, K. Li, J. Peng, T. Thebaud, L. M. Velazquez, J. Villalba, and N. Dehak, “Solospeech: Enhancing intelligibility and quality in target speech extraction through a cascaded generative pipeline,” *arXiv preprint arXiv:2505.19314*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.19314>

- [5] W. Wu, X. Chen, X. Wu, H. Li, and H. Meng, “Target speech extraction with pre-trained av-hubert and mask-and-recover strategy,” in *IJCNN* 2024.
- [6] W. Wu, X. Chen, S. Wang, J. Wang, L. Meng, X. Wu, H. Meng, and H. Li, “ c^2 av-tse: Context and confidence-aware audio visual target speaker extraction,” *IEEE Journal of Selected Topics in Signal Processing*, 2025.
- [7] J. Li, W. Wu, S. Wang, Z. Pan, K. A. Lee, H. Meng, and H. Li, “Memo: Attentional momentum for real-time audio-visual speaker extraction under impaired visual conditions,” *arXiv preprint arXiv:2507.15294*, 2025.
- [8] W. Wu, S. Wang, X. Wu, H. Meng, and H. Li, “Incorporating linguistic constraints from external knowledge source for audio-visual target speech extraction,” in *Interspeech* 2025.
- [9] W. Wenxuan, S. Wang, X. Wu, H. Meng, and H. Li, “ELEGANCE: Efficient LLM guidance for audio-visual target speech extraction,” *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 2801–2814, 2026.
- [10] B. Tang, B. Zeng, and M. Li, “Lauratse: Target speaker extraction using auto-regressive decoder-only language models,” *arXiv preprint arXiv:2504.07402*, 2025.
- [11] Z. Du, J. Wang, Q. Chen, Y. Chu, Z. Gao, Z. Li, K. Hu, X. Zhou, J. Xu, Z. Ma *et al.*, “Lauragt: Listen, attend, understand, and regenerate audio with gpt,” *arXiv preprint arXiv:2310.04673*, 2023.
- [12] B. Kang, X. Zhu, Z. Zhang, Z. Ye, M. Liu, Z. Wang, Y. Zhu, G. Ma, J. Chen, L. Xiao, C. Weng, W. Xue, and L. Xie, “LLaSE-g1: Incentivizing generalization capability for LLaMA-based speech enhancement,” Vienna, Austria: Association for Computational Linguistics, Jul. 2025.
- [13] W. Wu, X. Chen, X. Wu, H. Li, and H. Meng, “Avhumar: Audio-visual target speech extraction with pre-trained av-hubert and mask-and-recover strategy,” *CVPR 2024 Sight and Sound Workshop*.
- [14] Z. Pan, W. Wang, M. Borsdorf, and H. Li, “Imaginet: Target speaker extraction with intermittent visual cue through embedding inpainting,” in *ICASSP* 2023, 2023.
- [15] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “Roberta: A robustly optimized bert pretraining approach,” *ArXiv*, 2019.
- [16] Q. Team, “Qwen3 technical report,” 2025. [Online]. Available: <https://arxiv.org/abs/2505.09388>
- [17] J. Yao, H. Liu, C. CHEN, Y. Hu, E. Chng, and L. Xie, “Gense: Generative speech enhancement via language models using hierarchical modeling,” in *International Conference on Representation Learning*, Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, Eds., 2025.
- [18] H. Yang, J. Su, M. Kim, and Z. Jin, “Genhancer: High-fidelity speech enhancement via generative modeling on discrete codec tokens,” *Interspeech* 2024.
- [19] Z. Ye, P. Sun, J. Lei, H. Lin, X. Tan, Z. Dai, Q. Kong, J. Chen, J. Pan, Q. Liu *et al.*, “Codec does matter: Exploring the semantic shortcoming of codec for audio language model,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 24, 2025, pp. 25 697–25 705.
- [20] Z. Ye, X. Zhu, C.-M. Chan, X. Wang, X. Tan, J. Lei, Y. Peng, H. Liu, Y. Jin, Z. Dai *et al.*, “Llasa: Scaling train-time and inference-time compute for llama-based speech synthesis,” *arXiv preprint arXiv:2502.04128*, 2025.
- [21] J. Xu, Z. Guo, J. He, H. Hu, T. He, S. Bai, K. Chen, J. Wang, Y. Fan, K. Dang *et al.*, “Qwen2. 5-omni technical report,” *arXiv preprint arXiv:2503.20215*, 2025.
- [22] J. Xu, Z. Guo, H. Hu, Y. Chu, X. Wang, J. He, Y. Wang, X. Shi, T. He, X. Zhu *et al.*, “Qwen3-omni technical report,” *arXiv preprint arXiv:2509.17765*, 2025.
- [23] M. L. Team, B. Wang, B. Xiao, B. Zhang, B. Rong, B. Chen, C. Wan, C. Zhang, C. Huang, C. Chen *et al.*, “Longcat-flash-omni technical report,” *arXiv preprint arXiv:2511.00279*, 2025.
- [24] H. Ye, C.-H. H. Yang, A. Goel, W. Huang, L. Zhu, Y. Su, S. Lin, A.-C. Cheng, Z. Wan, J. Tian *et al.*, “Omnivinci: Enhancing architecture and data for omni-modal understanding llm,” *arXiv preprint arXiv:2510.15870*, 2025.
- [25] Z. Pan, X. Qian, and H. Li, “Speaker extraction with co-speech gestures cue,” *IEEE Signal Processing Letters*, vol. 29, pp. 1467–1471, 2022.
- [26] T. Afouras, J. S. Chung, A. Senior, O. Vinyals, and A. Zisserman, “Deep audio-visual speech recognition,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 12, pp. 8717–8727, 2022.
- [27] B. Shi, W.-N. Hsu, K. Lakhotia, and A. Mohamed, “Learning audio-visual speech representation by masked multimodal cluster prediction,” in *ICLR* 2024, 2024.

- [28] T. Saeki, S. Maiti, S. Takamichi, S. Watanabe, and H. Saruwatari, "SpeechBERTScore: Reference-aware automatic evaluation of speech generation leveraging nlp evaluation metrics," in *Interspeech 2024*.
- [29] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, M. Zeng, and F. Wei, "Wavlm: Large-scale self-supervised pre-training for full stack speech processing," *IEEE JSTSP 2021*.
- [30] G. Mittag, B. Naderi, A. Chehadi, and S. Möller, "Nisqa: A deep cnn-self-attention model for multidimensional speech quality prediction with crowdsourced datasets," in *Interspeech*, 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:233296150>
- [31] C. K. Reddy, V. Gopal, and R. Cutler, "Dnsmos p. 835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors," in *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2022, pp. 886–890.
- [32] J. S. Chung, A. Nagrani, and A. Zisserman, "Voxceleb2: Deep speaker recognition," in *INTERSPEECH*, 2018.
- [33] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower Provost, S. Kim, J. Chang, S. Lee, and S. Narayanan, "Iemocap: Interactive emotional dyadic motion capture database," *Language Resources and Evaluation*, vol. 42, pp. 335–359, 12 2008.
- [34] Z. Pan, M. Ge, and H. Li, "Usev: Universal speaker extraction with visual cue," *IEEE/ACM Trans. Audio, Speech and Lang. Proc.*, vol. 30, p. 3032–3045, sep 2022.
- [35] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [36] P.-J. Ku, A. H. Liu, R. Korostik, S.-F. Huang, S.-W. Fu, and A. Jukić, "Generative speech foundation model pretraining for high-quality speech extraction and restoration," in *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.
- [37] J. Lin, X. Cai, H. Dinkel, J. Chen, Z. Yan, Y. Wang, J. Zhang, Z. Wu, Y. Wang, and H. Meng, "Av-sepformer: Cross-attention sepformer for audio-visual target speaker extraction," in *ICASSP 2023*, 2023.
- [38] S. Lee, C. Jung, Y. Jang, J. Kim, and J. S. Chung, "Seeing through the conversation: Audio-visual speech separation based on diffusion model," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 12 632–12 636.
- [39] C. Jung, S. Lee, J. H. Kim, and J. S. Chung, "Flowavse: Efficient audio-visual speech enhancement with conditional flow matching," in *Interspeech*, 09 2024, pp. 2210–2214.